

CSE-221

Algorithms

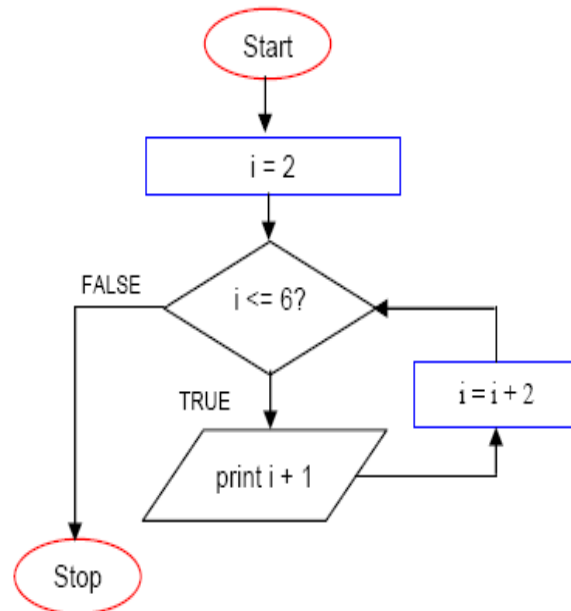
Introduction to Algorithms

Algorithm Definition

- A finite set of statements that guarantees an optimal solution in finite interval of time
- Algorithmic thinking and problem solving skill are vital in making efficient solutions.
- The English word "ALGORITHM" derives from the Latin word AL-KHWARIZMI'S name. He developed the concept of an algorithm in Mathematics, and thus sometimes being called the "Grandfather of Computer Science".

Glance of Algorithm

- An algorithm is a finite set of instructions or logic, written in order, to accomplish a certain predefined task.
- Algorithm is not the complete code or program
- Can be expressed either as an informal high level description as pseudocode or using a flowchart.



WHILE loop

- Do the loop body if the condition is true.
- Example: Get the sum of 1, 2, 3, ..., 100.
 - Algorithm:
 - Set the number = 1
 - Set the total = 0
 - While (number <= 100)
 - total = total + number
 - number = number + 1
 - End While
 - Display total

Algorithm Specifications

- *Input* - Every Algorithm must take zero or more number of input values from external.
- *Output* - Every Algorithm must produce an output as result.
- *Definiteness* - Every statement/instruction in an algorithm must be clear and unambiguous (only one interpretation)
- *Finiteness* - For all different cases, the algorithm must produce result within a finite number of steps.
- *Effectiveness* - Every Instruction must be basic enough to be carried out and it also must be feasible.

Good Algorithms?

- Run in less time
- Consume less memory

But computational resources (time complexity) usually important

Analyzing Algorithms

- Predict the amount of resources required:
 - **memory**: how much space is needed?
 - **computational time**: how fast the algorithm runs?
 - FACT: running time grows with the size of the input
 - Input size (number of elements in the input)
 - Size of an array, polynomial degree, # of elements in a matrix, # of bits in the binary representation of the input, vertices and edges in a graph
- Def: Running time = the number of primitive operations (steps) executed before termination*
- Arithmetic operations (+, -, *), data movement, control, decision making (*if, while*), comparison

Algorithm Analysis: Example

- *Alg.:* MIN ($a[1], \dots, a[n]$)

$m \leftarrow a[1];$

 for $i \leftarrow 2$ to n

 if $a[i] < m$

 then $m \leftarrow a[i];$

- **Running time:**

- the number of primitive operations (steps) executed before termination

$$T(n) = 1 \text{ [first step]} + (n) \text{ [for loop]} + (n-1) \text{ [if condition]} + (n-1) \text{ [the assignment in then]} = 3n - 1$$

- **Order (rate) of growth:**

- The leading term of the formula

- Expresses the asymptotic behavior of the algorithm

Typical Running Time Functions

- 1 (constant running time):
 - Instructions are executed once or a few times
- $\log N$ (logarithmic)
 - A big problem is solved by cutting the original problem in smaller sizes, by a constant fraction at each step
- N (linear)
 - A small amount of processing is done on each input element
- $N \log N$
 - A problem is solved by dividing it into smaller problems, solving them independently and combining the solution

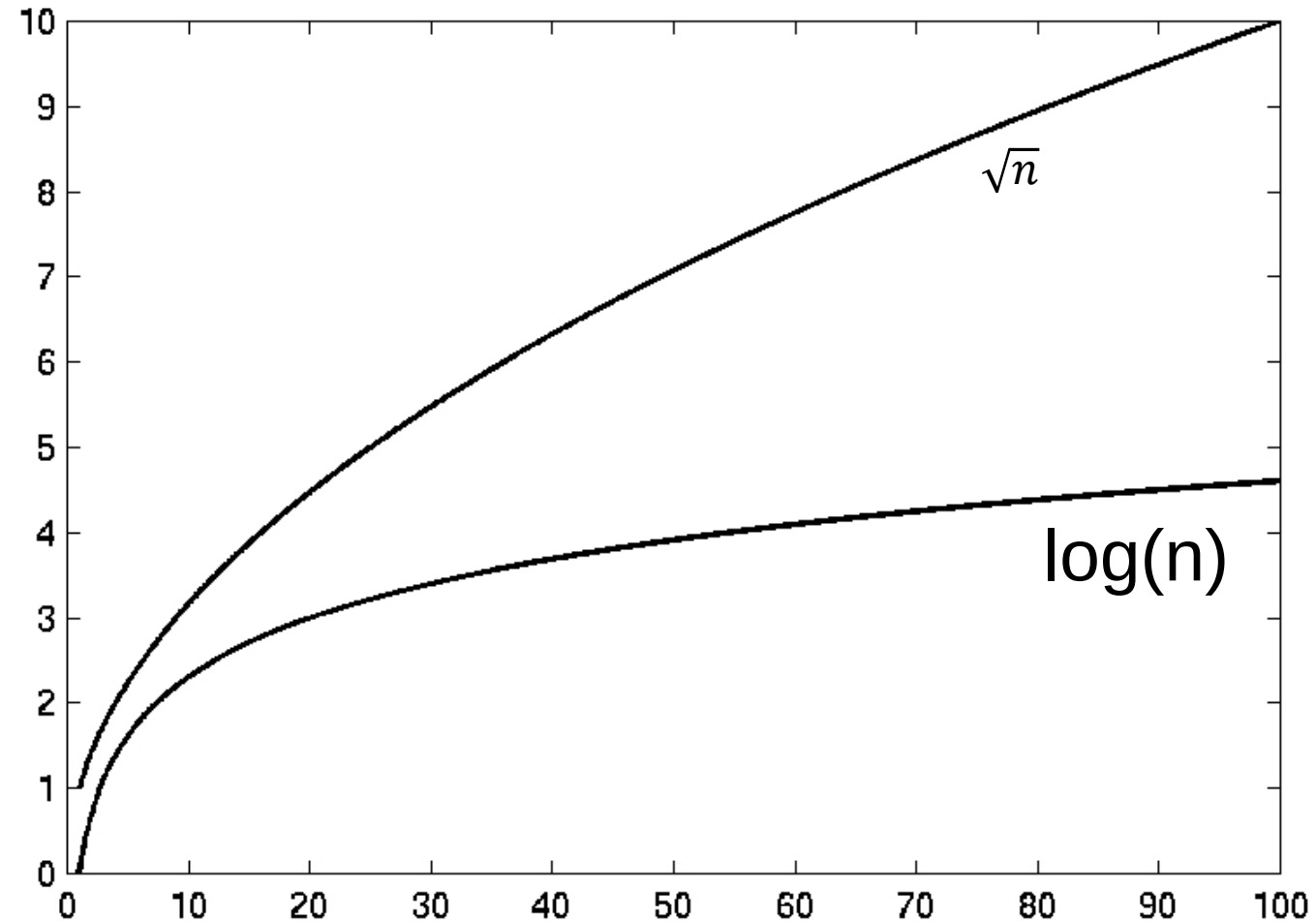
Typical Running Time Functions

- N^2 (quadratic)
 - Typical for algorithms that process all pairs of data items (double nested loops)
- N^3 (cubic)
 - Processing of triples of data (triple nested loops)
- N^K (polynomial)
- 2^N (exponential)
 - Few exponential algorithms are appropriate for practical use

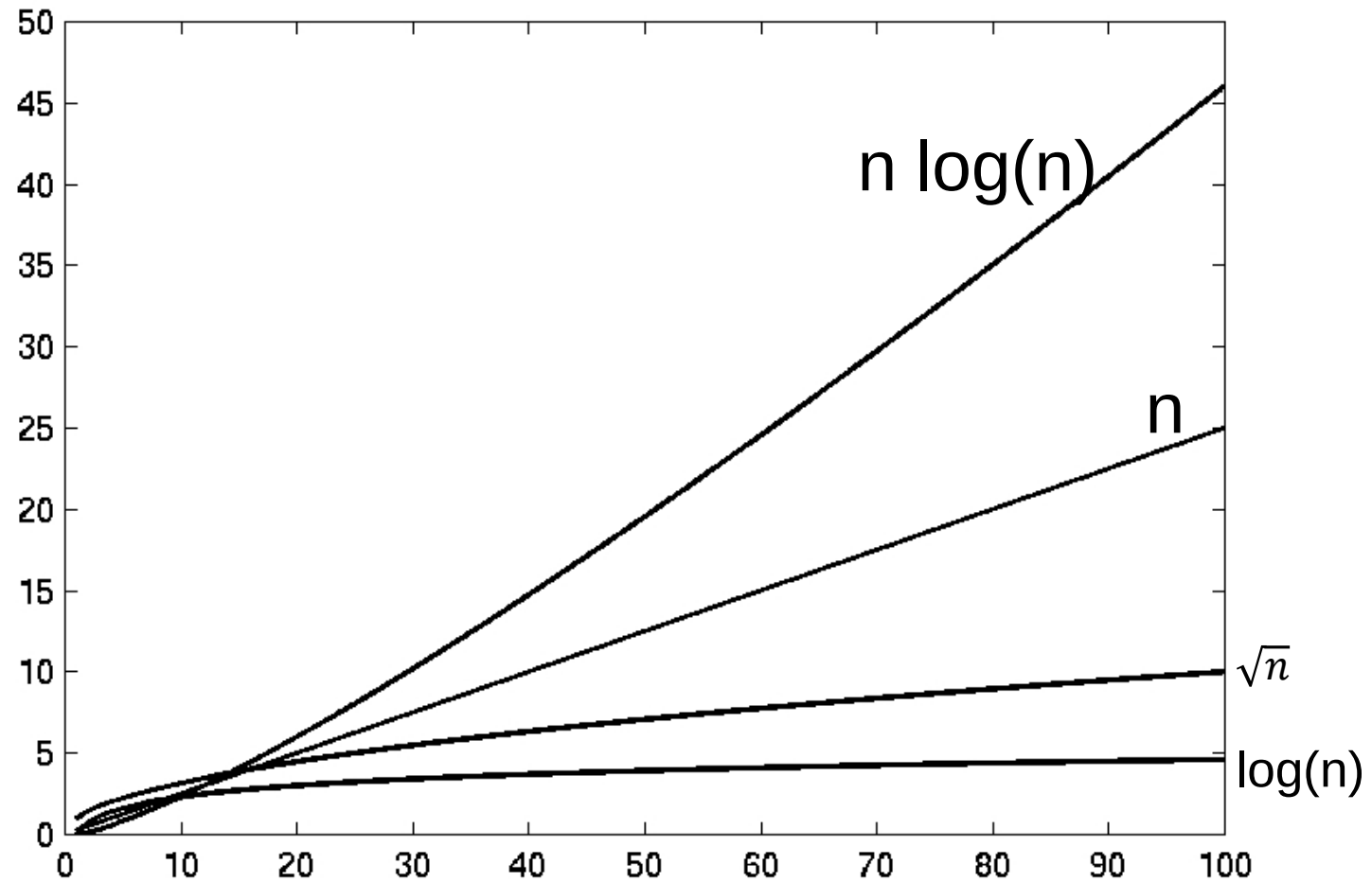
Growth of Functions

n	1	lgn	n	n lgn	n²	n³	2ⁿ
1	1	0.00	1	0	1	1	2
10	1	3.32	10	33	100	1,000	1024
100	1	6.64	100	664	10,000	1,000,000	1.2×10^{30}
1000	1	9.97	1000	9970	1,000,000	10^9	1.1×10^{301}

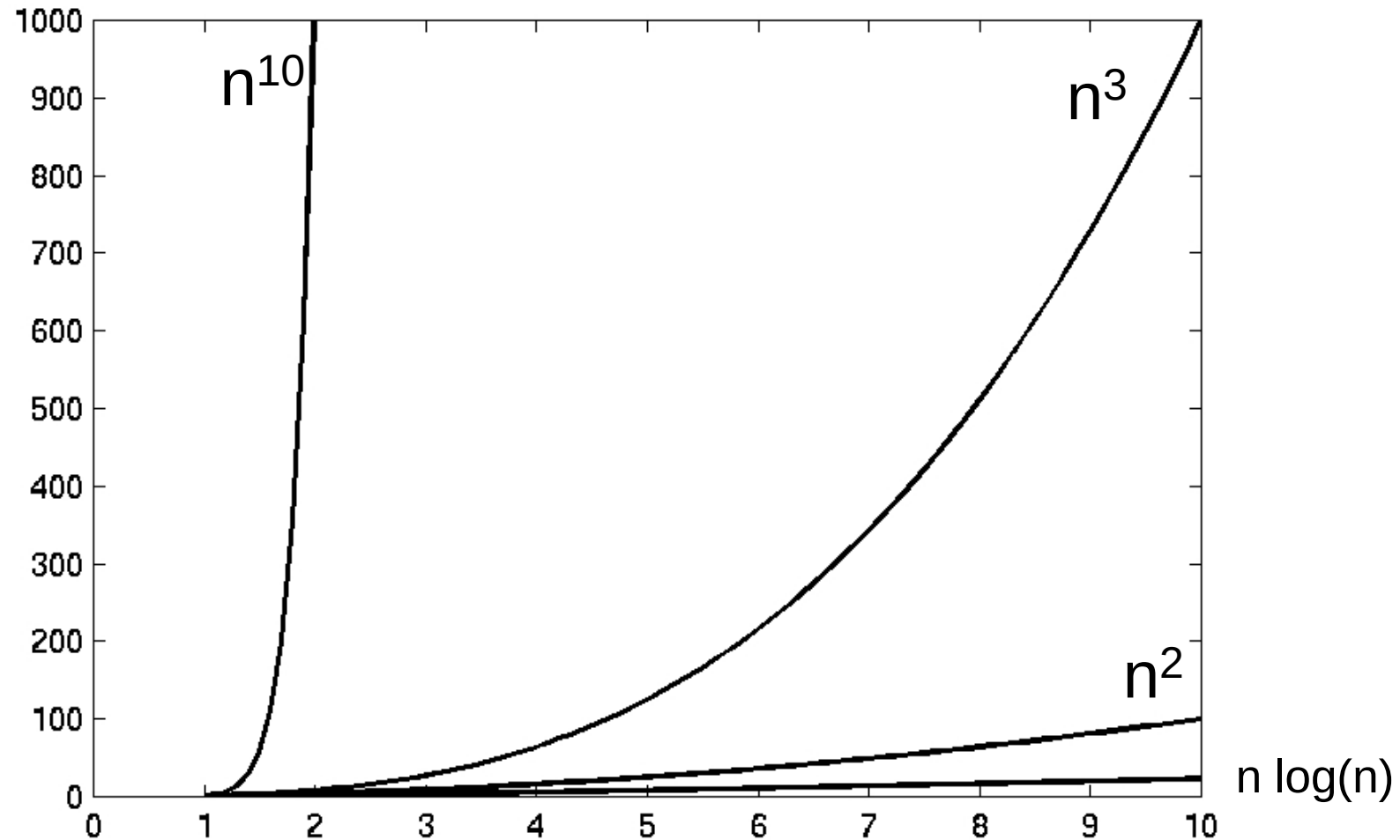
Complexity Graphs



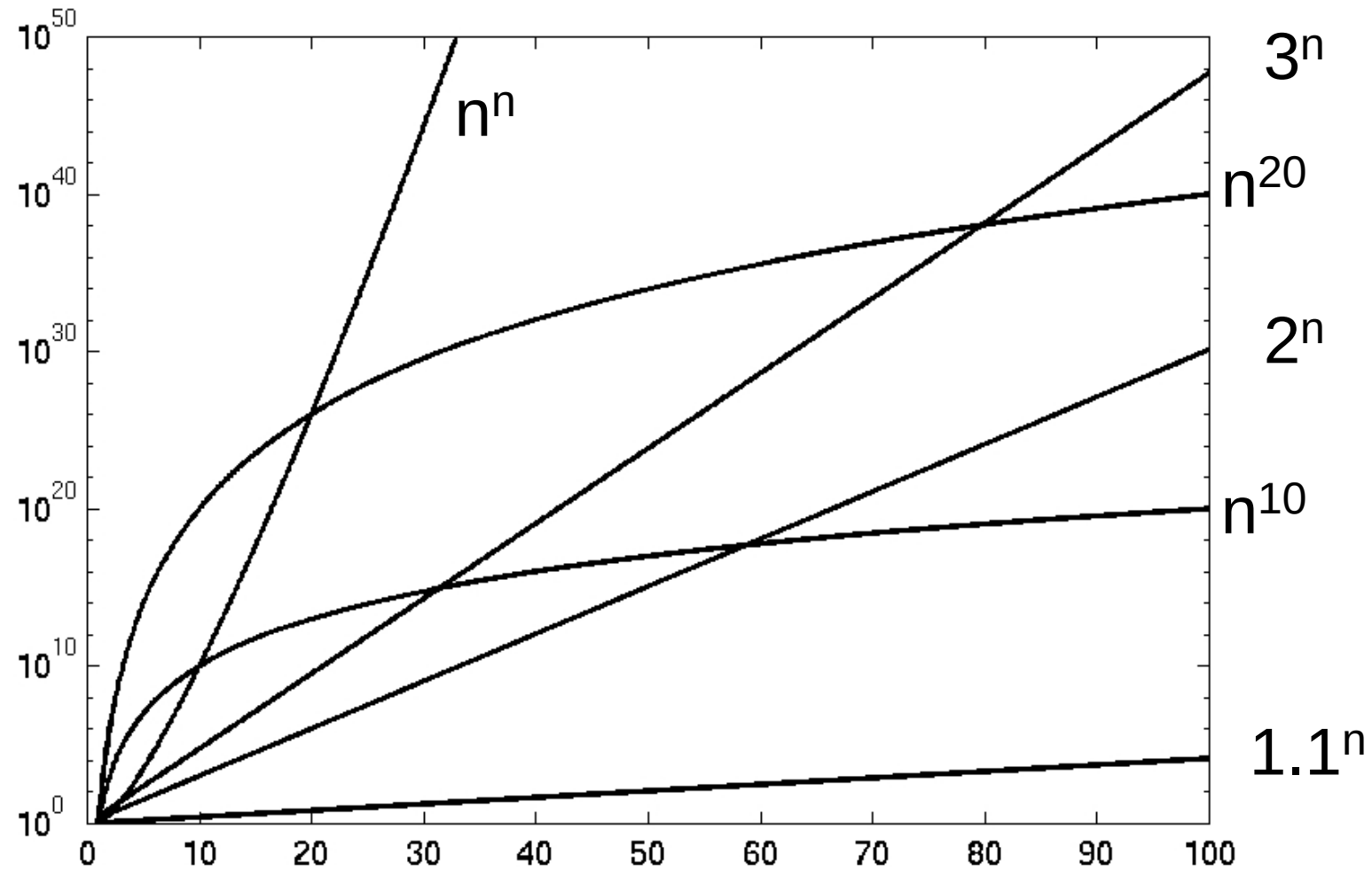
Complexity Graphs



Complexity Graphs



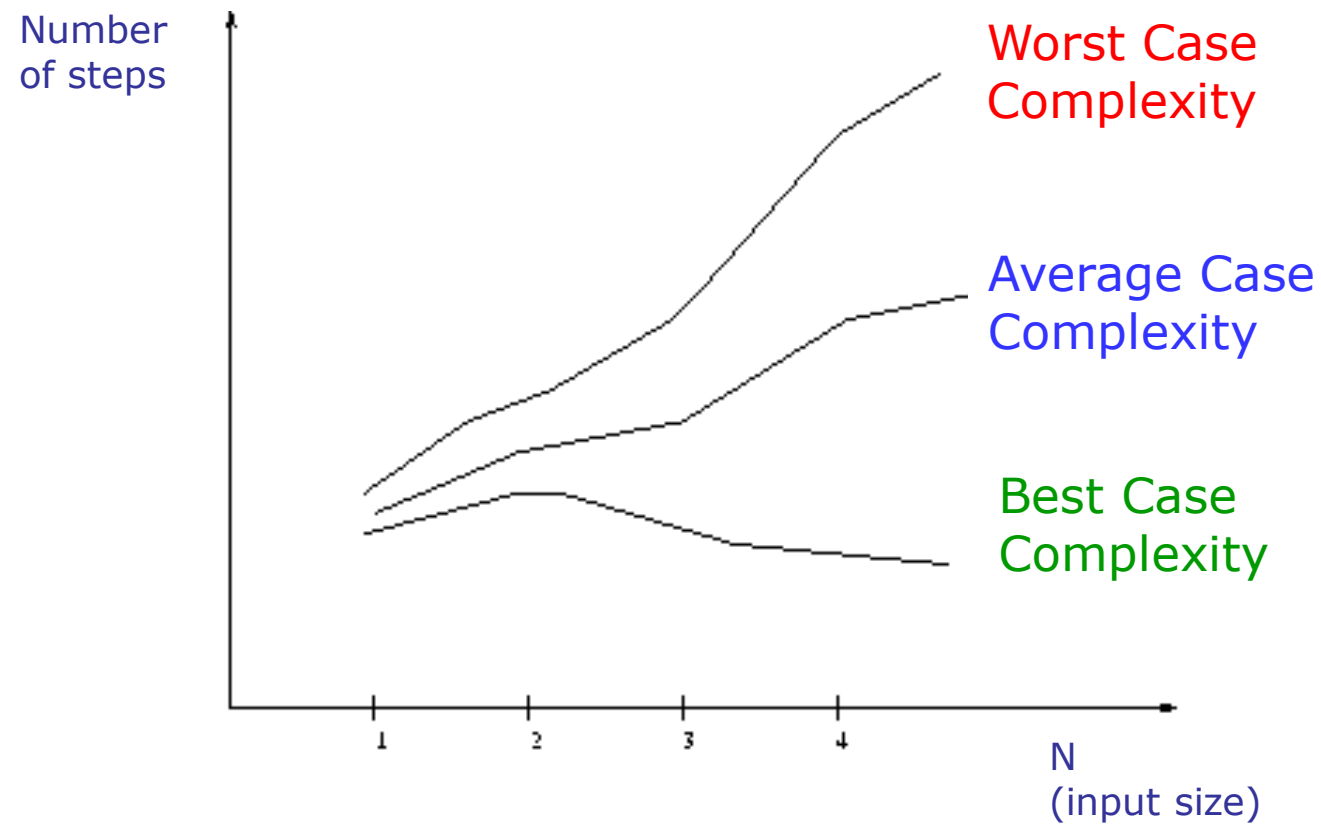
Complexity Graphs (log scale)



Algorithm Complexity

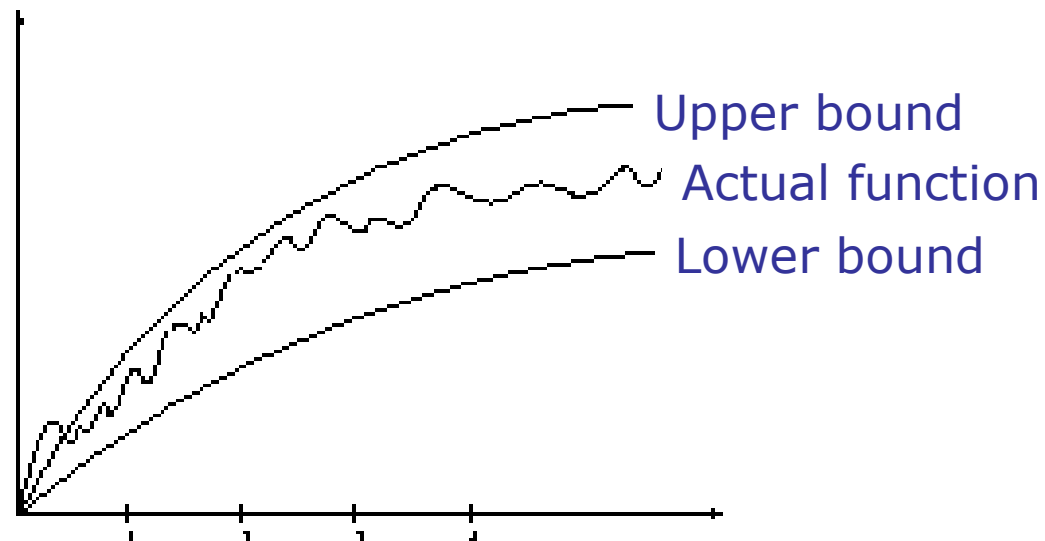
- **Worst Case Complexity:**
 - the function defined by the *maximum* number of steps taken on any instance of size n
- **Best Case Complexity:**
 - the function defined by the *minimum* number of steps taken on any instance of size n
- **Average Case Complexity:**
 - the function defined by the *average* number of steps taken on any instance of size n

Best, Worst, and Average Case Complexity



Doing the Analysis

- It's hard to estimate the running time exactly
 - Best case depends on the input
 - Average case is difficult to compute
 - So we usually focus on worst case analysis
 - Easier to compute
 - Usually close to the actual running time
- Strategy: find a function (an equation) that, for large n , is an upper bound to the actual function (actual number of steps, memory usage, etc.)



Motivation for Asymptotic Analysis

- *An exact computation* of worst-case running time can be difficult
 - Function may have many terms:
 - $4n^2 - 3n \log n + 17.5n - 43n^{2/3} + 75$
- *An exact computation* of worst-case running time is unnecessary

Classifying functions by their Asymptotic Growth Rates

- asymptotic growth rate, asymptotic order, or order of functions
 - Comparing and classifying functions that ignores
 - *constant factors* and
 - *small inputs*.
- The Sets big oh $O(g)$, big theta $\Theta(g)$, big omega $\Omega(g)$

Classifying functions by their Asymptotic Growth Rates

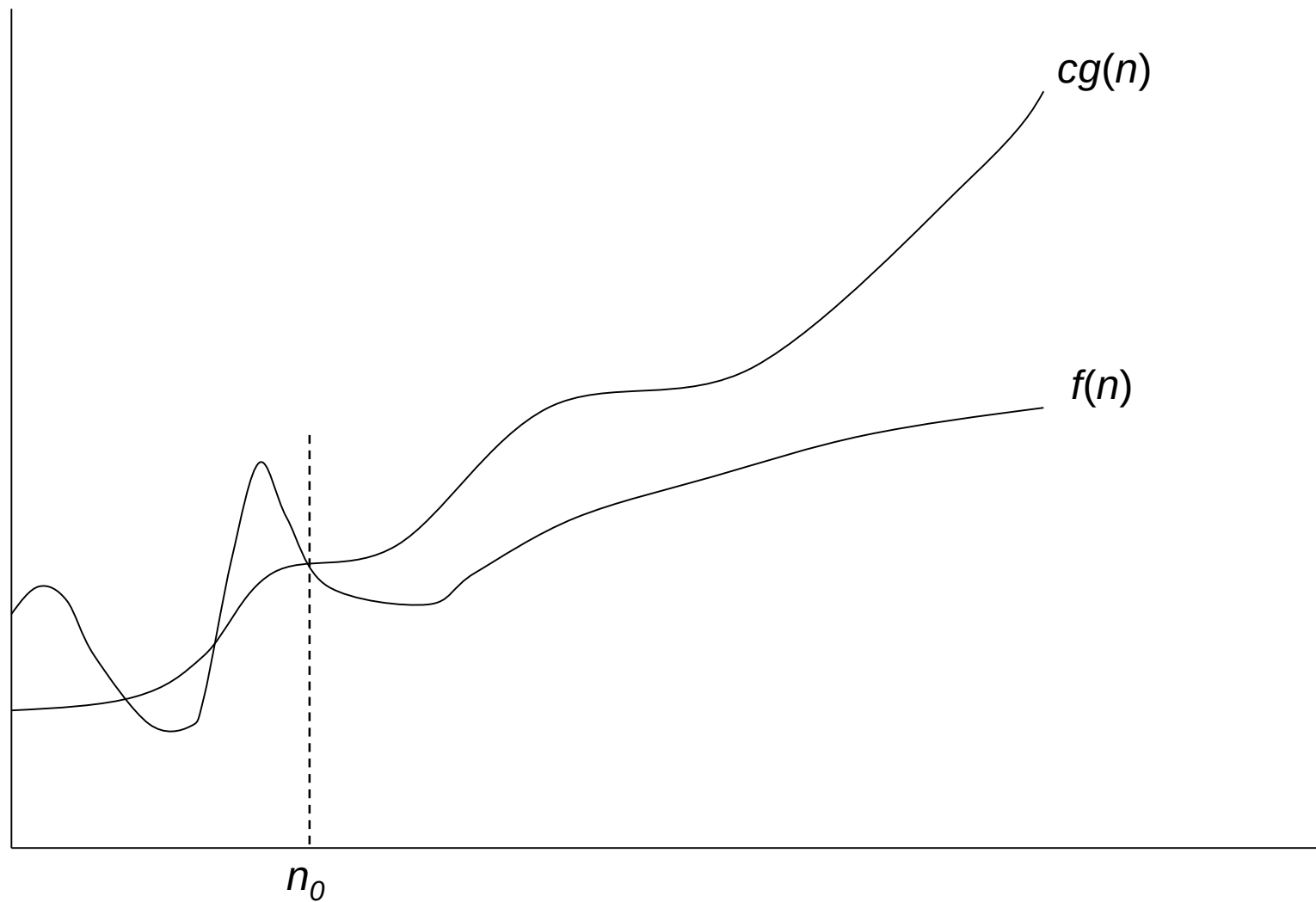
1. $O(g(n))$, Big-Oh of g of n , the Asymptotic Upper Bound
2. $\Theta(g(n))$, Theta of g of n , the Asymptotic Tight Bound
3. $\Omega(g(n))$, Omega of g of n , the Asymptotic Lower Bound

Big-O

$f(n) = O(g(n))$: there exist positive constants c and n_0 such that
 $0 \leq f(n) \leq cg(n)$ for all $n \geq n_0$

- What does it mean?
 - If $f(n) = O(n^2)$, then:
 - $f(n)$ can be larger than n^2 sometimes, **but...**
 - We can choose some constant **c** and some value n_0 such that for **every** value of **n** larger than n_0 : $f(n) < cn^2$
 - That is, for values larger than n_0 , $f(n)$ is never more than a constant multiplier greater than n^2
 - Or, in other words, $f(n)$ does not grow more than a constant factor faster than n^2 .

Visualization of $O(g(n))$



Examples

- $2n^2 = O(n^3)$: $2n^2 \leq cn^3 \Rightarrow 2 \leq cn \Rightarrow c = 1 \text{ and } n_0 = 2$

- $n^2 = O(n^2)$: $n^2 \leq cn^2 \Rightarrow c \geq 1 \Rightarrow c = 1 \text{ and } n_0 = 1$

- $1000n^2 + 1000n = O(n^2)$:

$$1000n^2 + 1000n \leq cn^2 \leq cn^2 + 1000n \Rightarrow c = 1001 \text{ and } n_0 = 1$$

- $n = O(n^2)$: $n \leq cn^2 \Rightarrow cn \geq 1 \Rightarrow c = 1 \text{ and } n_0 = 1$

Big-O

$$2n^2 = O(n^2)$$

$$1,000,000n^2 + 150,000 = O(n^2)$$

$$5n^2 + 7n + 20 = O(n^2)$$

$$2n^3 + 2 \neq O(n^2)$$

$$n^{2.1} \neq O(n^2)$$

More Big-O

$$20n^2 + 2n + 5 = O(n^2)$$

- Prove that:
- Let $c = 21$ and $n_0 = 4$
- $21n^2 > 20n^2 + 2n + 5$ for all $n > 4$
 $n^2 > 2n + 5$ for all $n > 4$

TRUE

Tight bounds

- We generally want the tightest bound we can find.
- While it is true that $n^2 + 7n$ is in $O(n^3)$, it is more interesting to say that it is in $O(n^2)$

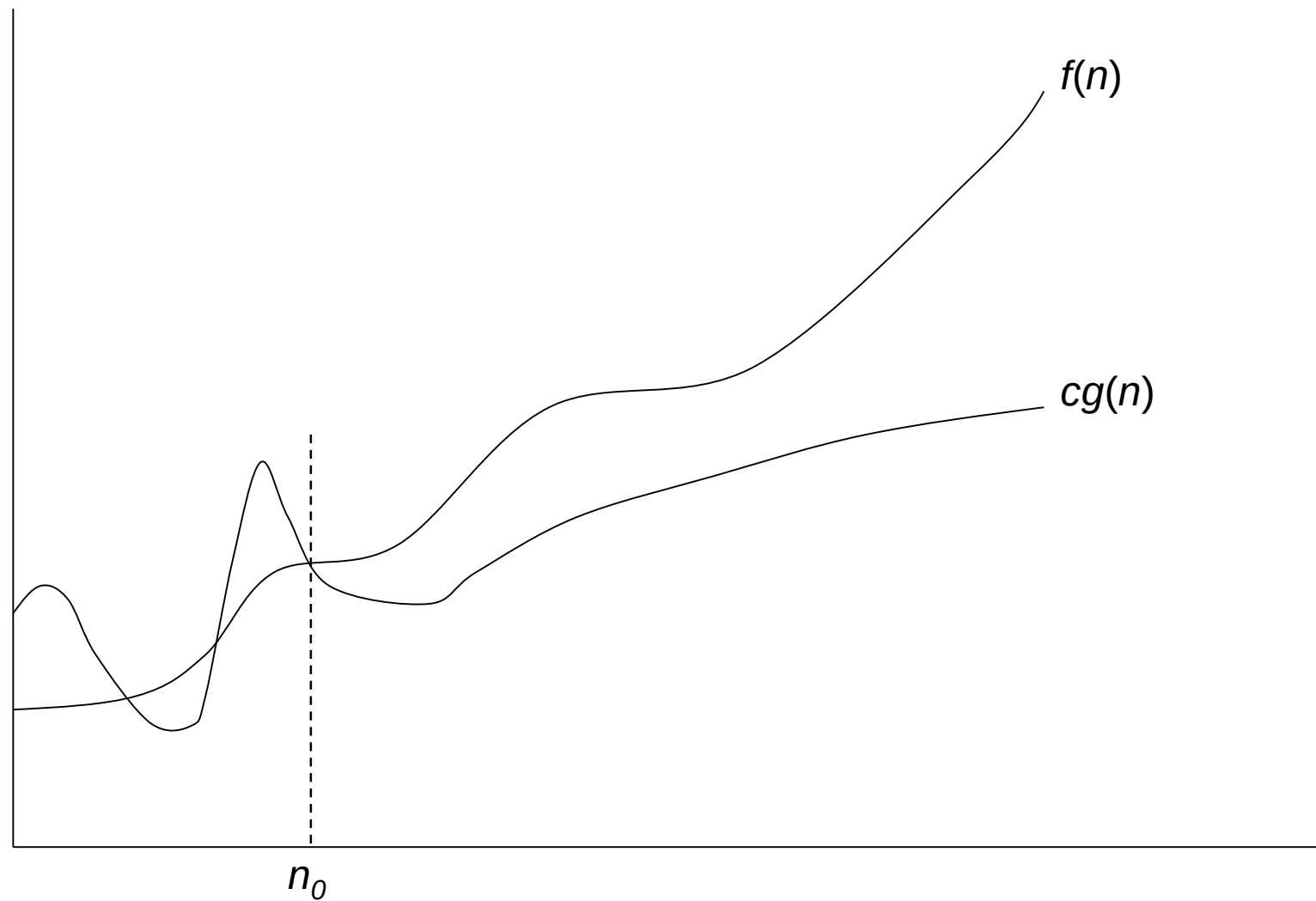
Big Omega – Notation

- $\Omega()$ – A **lower** bound

$f(n) = \Omega(g(n))$: there exist positive constants c and n_0 such that
 $0 \leq f(n) \geq cg(n)$ for all $n \geq n_0$

- $n^2 = \Omega(n)$
- Let $c = 1$, $n_0 = 2$
- For all $n \geq 2$, $n^2 > 1 \times n$

Visualization of $\Omega(g(n))$



Θ -notation

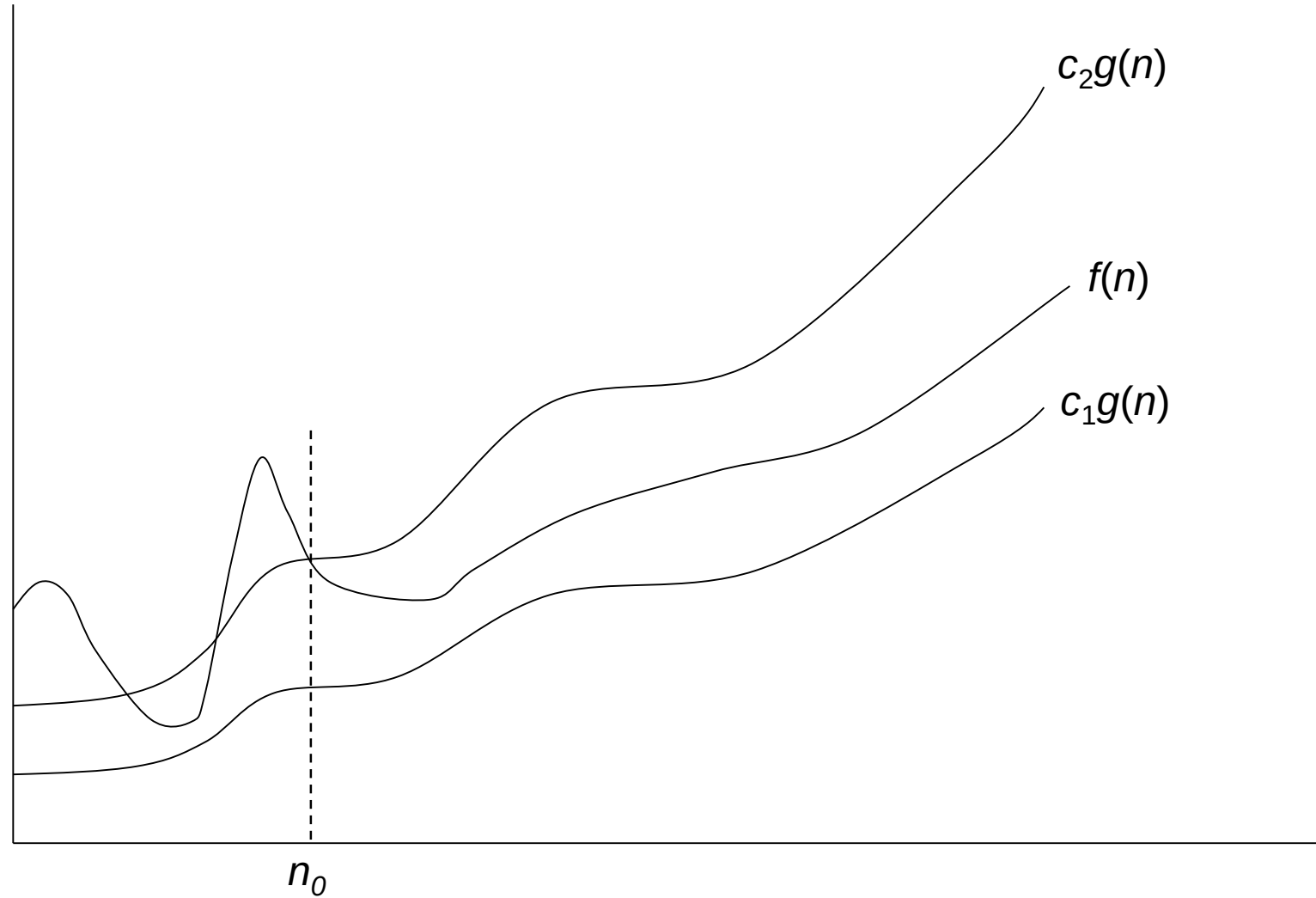
- Big-O is not a tight upper bound. In other words $n = O(n^2)$
- Θ provides a tight bound

$f(n) = \Theta(g(n))$: there exist positive constants c_1 , c_2 , and n_0 such that
 $0 \leq c_1 g(n) \leq f(n) \leq c_2 g(n)$ for all $n \geq n_0$

- In other words,

$$f(n) = \Theta(g(n)) \Rightarrow f(n) = O(g(n)) \text{ AND } f(n) = \Omega(g(n))$$

Visualization of $\Theta(g(n))$



A Few More Examples

- $n = O(n^2) \neq \Theta(n^2)$
- $200n^2 = O(n^2) = \Theta(n^2)$
- $n^{2.5} \neq O(n^2) \neq \Theta(n^2)$

Example 2

- Prove that: $20n^3 + 7n + 1000 = \Theta(n^3)$
- Let $c = 21$ and $n_0 = 10$
- $21n^3 > 20n^3 + 7n + 1000$ for all $n > 10$
 $n^3 > 7n + 5$ for all $n > 10$
TRUE, but we also need...
- Let $c = 20$ and $n_0 = 1$
- $20n^3 < 20n^3 + 7n + 1000$ for all $n \geq 1$
TRUE

Example 3

- Show that $2^n + n^2 = O(2^n)$

- Let $c = 2$ and $n_0 = 5$

$$2 \times 2^n > 2^n + n^2$$

$$2^{n+1} > 2^n + n^2$$

$$2^{n+1} - 2^n > n^2$$

$$2^n(2 - 1) > n^2$$

$$2^n > n^2 \quad \forall n \geq 5$$

✓

Asymptotic Notations - Examples

- Θ notation

- $n^2/2 - n/2 = \Theta(n^2)$
- $(6n^3 + 1)\lg n / (n + 1) = \Theta(n^2 \lg n)$
- n vs. n^2 $n \neq \Theta(n^2)$

- Ω notation

- n^3 vs. n^2 $n^3 = \Omega(n^2)$
- n vs. $\log n$ $n = \Omega(\log n)$
- n vs. n^2 $n \neq \Omega(n^2)$

- O notation

- $2n^2$ vs. n^3 $2n^2 = O(n^3)$
- n^2 vs. n^2 $n^2 = O(n^2)$
- n^3 vs. $n \log n$ $n^3 \neq O(n \lg n)$

Asymptotic Notations - Examples

- For each of the following pairs of functions, either $f(n)$ is $O(g(n))$, $f(n)$ is $\Omega(g(n))$, or $f(n) = \Theta(g(n))$. Determine which relationship is correct.

- $f(n) = \log n^2$; $g(n) = \log n + 5$

$f(n) = \Theta(g(n))$

- $f(n) = n$; $g(n) = \log n^2$

$f(n) = \Omega(g(n))$

- $f(n) = \log \log n$; $g(n) = \log n$

$f(n) = O(g(n))$

- $f(n) = n$; $g(n) = \log^2 n$

$f(n) = \Omega(g(n))$

- $f(n) = n \log n + n$; $g(n) = \log n$

$f(n) = \Omega(g(n))$

- $f(n) = 10$; $g(n) = \log 10$

$f(n) = \Theta(g(n))$

- $f(n) = 2^n$; $g(n) = 10n^2$

$f(n) = \Omega(g(n))$

- $f(n) = 2^n$; $g(n) = 3^n$

$f(n) = O(g(n))$

Simplifying Assumptions

1. If $f(n) = O(g(n))$ and $g(n) = O(h(n))$, then $f(n) = O(h(n))$
2. If $f(n) = O(kg(n))$ for any $k > 0$, then $f(n) = O(g(n))$
3. If $f_1(n) = O(g_1(n))$ and $f_2(n) = O(g_2(n))$,
then $f_1(n) + f_2(n) = O(\max(g_1(n), g_2(n)))$
4. If $f_1(n) = O(g_1(n))$ and $f_2(n) = O(g_2(n))$,
then $f_1(n) * f_2(n) = O(g_1(n) * g_2(n))$

Some Simplified Rules

- $O(1) = c$, where c is a constant
- $O(n) = c*n = cn$, where c is constant and n is variable
- $c_1*O(1) = c_1*c = c_2 = O(1)$, where c, c_1, c_2 are constants
 - $O(1) + O(1) + O(1) = 3*O(1) = O(1)$
 - $5*O(1) = O(1)$
- $n*O(1) = n*c = cn = O(n)$, where c is constant and n is variable
- $O(m) + O(n) \neq O(m+n)$
- $O(m) * O(n) = c_1mc_2n = (c_1*c_2)(mn) = (c_2)(mn) = O(mn)$
- $O(m)*O(n)*O(p)*O(q) = O(m(n(p(q)))) = O(mnpq)$
 - Example nested for loops
- $O(an^2 + bn + c) = O(n^2)$ where a, b, c are constants

Example #1: carry n books from one bookshelf to another one

- How many operations?
- n pick-ups, n forward moves, n drops and n reverse moves $\rightarrow 4n$ operations
- $4n$ operations = $c \cdot n = O(c \cdot n) = O(n)$
- Similarly, any program that reads n inputs from the user will have minimum time complexity $O(n)$.

Example #2: Locating Roll-Number record in Attendance Sheet

What is the time complexity of search?

- Binary Search algorithm at work
 - $O(\log n)$
- Sequential search?
 - $O(n)$

Example #3: Teacher of CSE 221 gives gifts to first 10 students

- There are n students in the queue.
- Teacher brings one gift at a time.
- Time complexity = $O(c \cdot 10) = O(1)$
- Teacher will take exactly same time irrespective of the line length.

Loops with Break

```
for (j = 0; j < n; ++j) {  
    // 3 atomics  
    if (condition) break;  
}
```

- Upper bound = $O(4n) = O(n)$
- Lower bound = $\Omega(4) = \Omega(1)$
- Complexity = $O(n)$

Ques: Why don't we have a $\Theta(\dots)$ notation here?

Sequential Search

- Given an **unsorted** vector/list $a[]$, find the location of element X .

```
for (i = 0; i < n; i++) {  
    if (a[i] == X) return true;  
}  
return false;
```

- Input size: $n = \text{array size}()$
- Complexity = $O(n)$

If-then-else Statement

```
if(condition)
    i = 0;
else
    for ( j = 0; j < n; j++)
        a[j] = j;
```

- Complexity = ??
 $= O(1) + \max (O(1), O(N))$
 $= O(1) + O(N)$
 $= O(N)$

Consecutive Statements

```
for (j = 0; j < n; ++j) {  
    // 3 atomics  
}  
for (j = 0; j < n; ++j) {  
    // 5 atomics  
}
```

- Add the complexity of consecutive statements
- Complexity = $O(3n + 5n) = O(n)$

Nested Loop Statements

- Analyze such statements inside out

```
for (j = 0; j < n; ++j) {  
    // 2 atomics  
    for (k = 0; k < n; ++k) {  
        // 3 atomics  
    }  
}
```

- Complexity = $O((2 + 3n)n) = O(n^2)$

Example

- Code:
- `a = b;`
- Complexity:

Example

- Code:

- ```
sum = 0;
```
- ```
for (i=1; i <=n; i++)
```
- ```
 sum += n;
```

- Complexity:

# Example

- Code:

- `sum = 0;`
- `for (j=1; j<=n; j++)`
- `for (i=1; i<=j; i++)`
- `sum++;`
- `for (k=0; k<n; k++)`
- `A[k] = k;`

- Complexity:



# Example

- Code:

```
sum1 = 0;
for (i=1; i<=n; i++)
 for (j=1; j<=n; j++)
 sum1++;
```
- Complexity:

# Example

- Code:
  - `sum2 = 0;`
  - `for (i=1; i<=n; i++)`
  - `for (j=1; j<=i; j++)`
  - `sum2++;`
- Complexity:

# Example

- Code:

```
sum1 = 0;
for (k=1; k<=n; k*=2)
 for (j=1; j<=n; j++)
 sum1++;
```
- Complexity:

# Example

- Code:
  - `sum2 = 0;`
  - `for (k=1; k<=n; k*=2)`
  - `for (j=1; j<=k; j++)`
  - `sum2++;`
- Complexity:

# Recursion

```
long factorial(int n)
{
 if(n <= 1)
 return 1;
 else
 return n*factorial(n- 1);
}
```

In terms of big-Oh:

$$t(1) = 1$$

$$t(n) = 1 + t(n-1) = 1 + 1 + t(n-2)$$

$$= \dots k + t(n-k)$$

Choose  $k = n-1$

$$t(n) = n-1 + t(1) = n-1 + 1 = O(n)$$

Consider the following time complexity:

$$t(0) = 1$$

$$t(n) = 1 + 2t(n-1) = 1 + 2(1 + 2t(n-2)) = 1 + 2 + 4t(n-2)$$

$$= 1 + 2 + 4(1 + 2t(n-3)) = 1 + 2 + 4 + 8t(n-3)$$

$$= 1 + 2 + \dots + 2^{k-1} + 2^k t(n-k)$$

Choose  $k = n$

$$t(n) = 1 + 2 + \dots + 2^{n-1} + 2^n = 2^{n+1} - 1$$

# Binary Search

- Given a **sorted** vector/list `a[ ]`, find the location of element `X`

```
unsigned int binary_search(vector<int> a, int X)
{
 unsigned int low = 0, high = a.size()-1;

 while (low <= high) {
 int mid = (low + high) / 2;
 if (a[mid] < X)
 low = mid + 1;
 else if(a[mid] > X)
 high = mid - 1;
 else
 return mid;
 }
 return NOT_FOUND;
}
```

- Input size:  $n = \text{array size}()$
- Complexity =  $O(k \text{ iterations} \times (1 \text{ comparison} + 1 \text{ assignment}) \text{ per loop})$   
=  $O(\log(n))$

# Summary

- Time complexity is a measure of algorithm efficiency
- Efficient algorithm plays the major role in determining the running time.

**Q: Is it possible to determine running time based on algorithm's time complexity alone?**

- Minor tweaks in the code can cut down the running time by a factor too.
- Other items like CPU speed, memory speed, device I/O speed can help as well.
- For certain problems, it is possible to allocate additional space & improve time complexity.

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